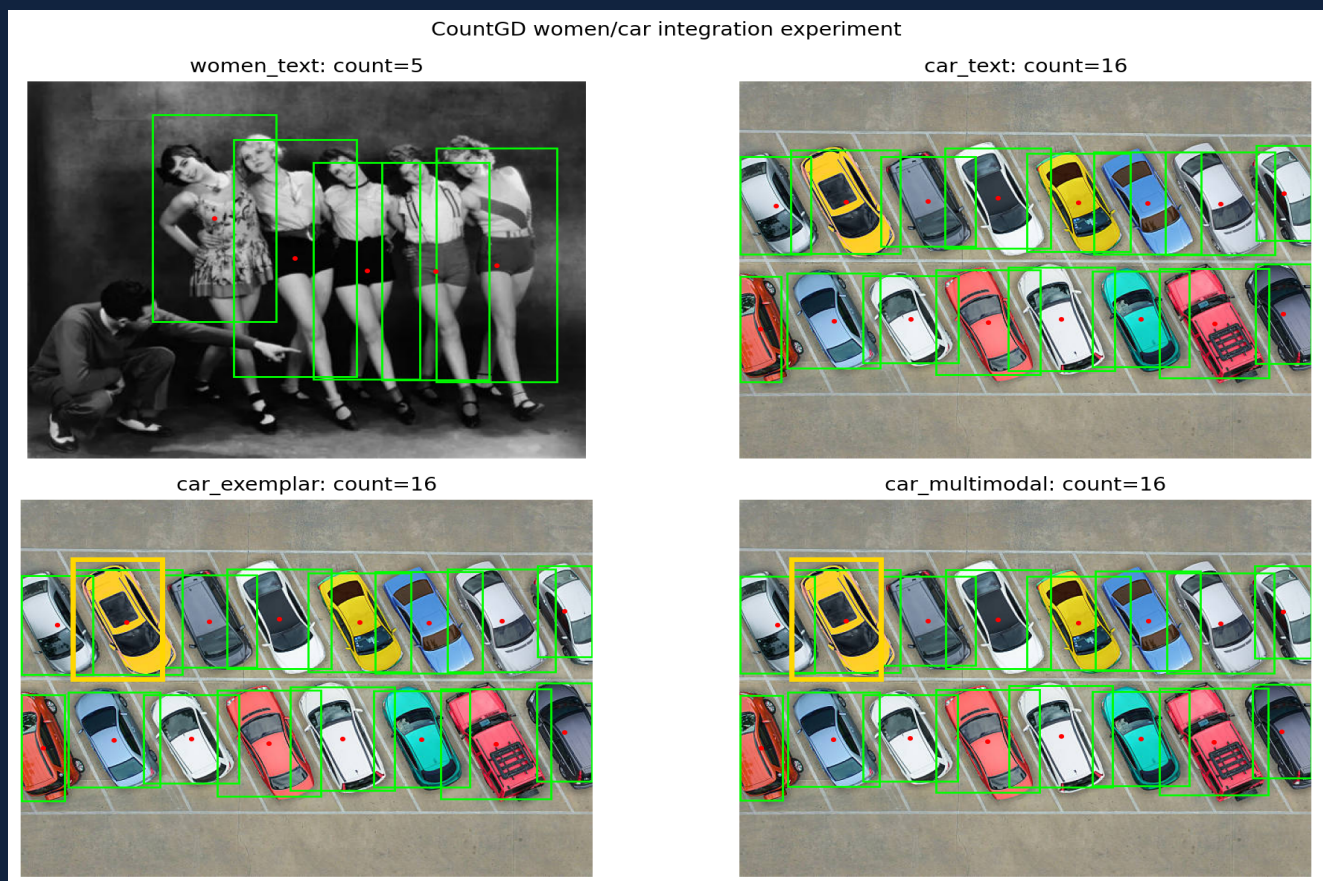


CountGD Component Analysis and Open-World Counting

Team Project 1: image encoder, text encoder, ROIAlign, similarity matrix, top-K selection, and integrated experiments



TEAM SIZE Up to 5 students

SUBMIT notebook code, research report PDF, and oral-presentation PPTX

Project workflow

This assignment is designed for teams of up to five. Assign one primary owner to each component: image encoder, text encoder, ROIAlign, similarity matrix, and top-K selection. Smaller teams combine roles. The whole team remains responsible for integration, experimental validity, the report, and the oral defence.

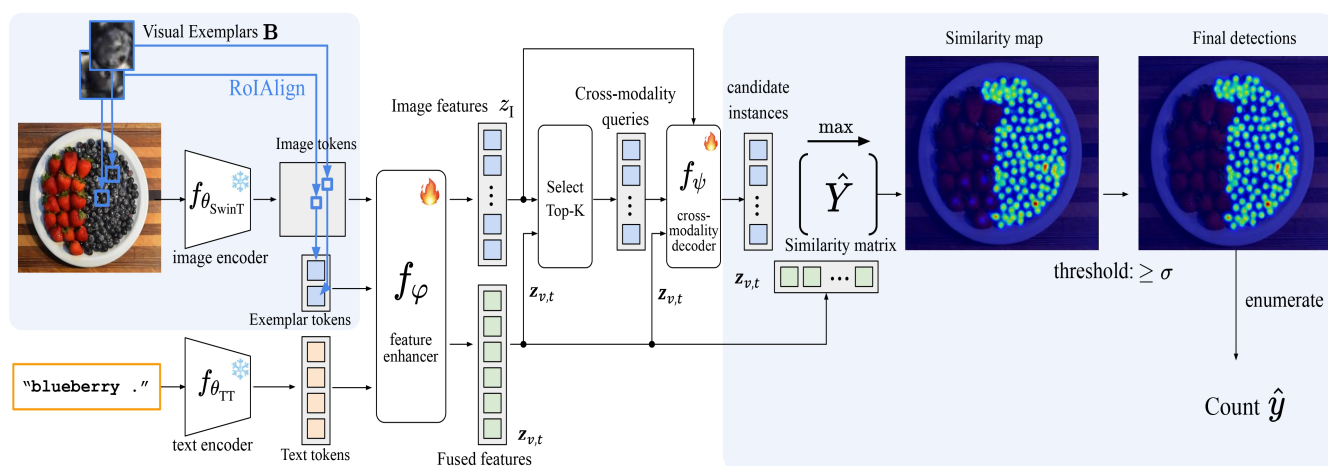
Role	Primary component	Required evidence
A	Image encoder	multi-scale/combined feature shapes and activation
B	Text encoder	token IDs, masks, and projected token shapes
C	ROIAlign	box coordinates and exemplar-token output
D	Similarity matrix	matrix shape, mask, and heatmap
E	Top-K	ranked proposals, prompt IDs, and interpretation

Test	Prompt	Expected count
women.jpg	text: women	5
car.jpg	text: car	16
car.jpg	one yellow-car exemplar	16
car.jpg	text plus exemplar	16

Runtime setup

- Use Python 3.9 and install requirements.txt.
- The official source snapshot is included under vendor/countgd/.
- Install the checkpoint and BERT assets using checkpoints/README.md, or set COUNTGD_CHECKPOINT and COUNTGD_BERT.
- Run SC5006_CountGD_Project_Student.ipynb in order and complete all five component TODOs.
- Use the official checkpoint for final evidence; label any cached output clearly.

Model connection



The assessed operations connect directly to official CountGD source: ROIAlign creates visual exemplar tokens, ContrastiveEmbed forms query-prompt scores, and encoder top-K selects proposals before decoder prediction.

Tasks 1-6 - CountGD Component Analysis

Complete every component TODO and connect the evidence in the shared integration experiment.

Task 1 - Image encoder (15%)

Run CountGD's nested-tensor and Swin-backbone path, preserve multi-scale outputs, and construct the combined 1/8-scale feature map.

TODOs Report input and feature-map shapes. | Visualize mean activation. | Explain multi-scale evidence.

Task 2 - Text encoder (15%)

Reproduce caption tokenization, special-token attention masks, BERT encoding, and projection into the shared hidden dimension.

TODOs Return prompt tokens and valid-token mask. | Identify real/padded columns. | Explain padding masks.

Task 3 - ROIAlign exemplar extraction (15%)

Extract 1x1 visual exemplar tokens from the combined feature map using resized pixel-space xyxy boxes.

TODOs Use `spatial_scale=1/8` and `aligned=True`. | Return (B,N,D) tokens. | Show the normalized-box error.

Task 4 - Similarity matrix (15%)

Calculate dot-product logits between query/image tokens and text/exemplar prompt tokens, then mask invalid prompt columns.

TODOs Validate (B,N,D), (B,T,D), (B,T). | Return (B,N,T). | Visualize and interpret rows/columns.

Task 5 - Top-K proposal selection (15%)

Reduce each proposal row to its strongest prompt score and select the K highest-scoring proposal rows.

TODOs Return scores, proposal IDs, prompt IDs. | Verify descending order. | Explain why top-K is not duplicate removal.

Task 6 - Integration and experiments (25%)

Run the official model on `women.jpg` and `car.jpg`, post-process predictions, and connect component evidence to counting performance.

TODOs Compare three prompt modes. | Add an ablation/sensitivity study. | Report runtime, hardware, and one failure case.

Submission and self-check

Submit exactly three team files: TeamXX_SC5006_Project1.ipynb, TeamXX_SC5006_Project1_Report.pdf, and TeamXX_SC5006_Project1_Presentation.pptx.

Report requirements

- 6-8 pages excluding references/appendix.
- Sections: Introduction, Related Work, Method, Experiments, Results and Discussion, Conclusion and Limitations, References, and team contributions.
- Include equations or pseudocode for ROIAlign, similarity, and top-K; add architecture/results figures, quantitative results, runtime, ablation, and failure analysis.

Oral presentation

- 10 minutes plus Q&A; 8-10 content slides.
- Every member must present their component and answer questions about the integrated pipeline.

Required checks

- No instructor-specific absolute path appears in notebook source.
- Official transform resizes women to (3,800,1041) and car to (3,800,1066).
- Checkpoint loading reports zero missing and unexpected keys.
- Every live forward pass returns (900,256) confidences and (900,4) boxes.
- At threshold 0.23 the four counts are 5, 16, 16, and 16.
- Student and solution notebooks have matching task order and interfaces.
- The contribution table identifies all members and their integration work.

References

Amini-Naieni, Han, and Zisserman. *CountGD: Multi-Modal Open-World Counting*. NeurIPS 2024.

Official code: github.com/niki-amini-naieni/CountGD

Project page: robots.ox.ac.uk/~vgg/research/countgd/